Frage **2**

Bisher nicht beantwortet

v1 (neueste)

Which gender do you identify with?

- male
- female
- diverse
- no specification

Frage **3**

Bisher nicht beantwortet

v3 (neueste)

Have you attended the lectures in which we talked about collective communication in MPI and performance analysis

- Yes, I have attended all lectures.
- Yes, I have attended some of the lectures.
- I didn’t attend the lectures, but I made sure to thoroughly review the material on my own
- I am currently not in Fulda and therefore cannot attend the lectures.
- I don't want to answer this question.

Frage **4**

Bisher nicht beantwortet

v1 (neueste)

Which semester of your Master's program are you in?

Antwort:

Frage **5**

Bisher nicht beantwortet

v1 (neueste)

Where did you grow up (country)?

Antwort:

Frage **6**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I haven't yet fully understood the topic of collective communication, which was the subject of the last lecture. I find it rather difficult to develop a program with collective MPI functions.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Information

v1 (neueste)

For the user test, please download the following two .c programs from Moodle:

- broadcast_p2p.c
- broadcast_coll.c

Open the two programs locally on your computer and run them with a different number of processes.

What happens in the programs? Where does the code differ?

Before you continue with the user test, make sure that you have fully understood the behavior of the two programs.

Frage **7**

Bisher nicht beantwortet

v1 (neueste)

Describe what the two programs do and what has been implemented within the two programs (without EduMPI analysis).

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

p 0 Wörter Build with tinyMCE

Frage **8**

Bisher nicht beantwortet

v2 (neueste)

Were you able to execute and understand the broadcast programs independently or did you need help?

- [ ] I was able to solve the tasks independently.
- [ ] I asked the person sitting next to me or another student for help.
- [ ] I have asked the teacher for help.
- [ ] I looked for help on the Internet or asked ChatGPT.
- [ ] I couldn't do the tasks at all.

Frage **9**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I have fully understood the code of the two given programs. I know what the programs do.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Information

v5 (neueste)

Now the two programs are to be compared with EduMPI. To interpret the results, remember the last user test and what was discussed in the lecture.

Before you start with the tasks, first read through the entire task.

1. Make sure that you have downloaded and are using the latest **EduMPI version 1.1.4**. Earlier versions can be removed.

2. The two programs have already been executed using 416 ranks so that not all students have to run large tests on the cluster simultaneously and there may not be enough time to complete the user test.
Load the two programs with the job IDs: 12322 (broadcast_p2p.c) and 12321 (broadcast_coll.c) from the database.

3. Now analyze and compare the two programs with the available options (views, runtime options, etc.) in EduMPI.

4. Answer the following questions.

Note: With EduMPI you can observe which point-to-point operations are carried out internally in MPI for collective functions.

Frage **10**

Bisher nicht beantwortet

v1 (neueste)

Which of the two programs is more efficient?

- broadcast_p2p.c
- broadcast_coll.c

Frage **11**

Bisher nicht beantwortet

v1 (neueste)

Shouldn't the runtime be the same if both programs perform a broadcast using the same number of processes and the same array? Try to explain why one of the two programs has a much longer runtime.

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

p 0 Wörter Build with tinyMCE

Maximale Dateigröße: 20 MB

Dateien

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Akzeptierte Dateitypen

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Frage **12**

Bisher nicht beantwortet

v1 (neueste)

Which of the two programs contains the typical problem of a bottleneck and why? Which rank represents the “bottleneck” here?
Please attach screenshots if you can identify the problem using a visualization in EduMPI.

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

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Frage **13**

Bisher nicht beantwortet

v1 (neueste)

Which of the two broadcast approaches offers better load balancing in an MPI program and why? How do you determine this with EduMPI? Please add screenshots.

Bearbeiten Ansicht Einfügen Format

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Frage **14**

Bisher nicht beantwortet

v1 (neueste)

If you have to distribute a message to all ranks for an MPI program in the future (broadcast), which of the two approaches (collective, p2p) would you choose and why?

Bearbeiten Ansicht Einfügen Format

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◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Information

v5 (neueste)

Now the two programs are to be compared with EduMPI. To interpret the results, remember the last user test and what was discussed in the lecture.

Before you start with the tasks, first read through the entire task.

1. Make sure that you have downloaded and are using the latest **EduMPI version 1.1.4**. Earlier versions can be removed.

2. The two programs have already been executed using 416 ranks so that not all students have to run large tests on the cluster simultaneously and there may not be enough time to complete the user test.
Load the two programs with the job IDs: 12322 (broadcast_p2p.c) and 12321 (broadcast_coll.c) from the database.

3. Now analyze and compare the two programs with the available options (views, runtime options, etc.) in EduMPI.

4. Answer the following questions.

Note: With EduMPI you can observe which point-to-point operations are carried out internally in MPI for collective functions.

Frage **10**

Bisher nicht beantwortet

v1 (neueste)

Which of the two programs is more efficient?

- broadcast_p2p.c
- broadcast_coll.c

Frage **11**

Bisher nicht beantwortet

v1 (neueste)

Shouldn't the runtime be the same if both programs perform a broadcast using the same number of processes and the same array? Try to explain why one of the two programs has a much longer runtime.

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

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Frage **12**

Bisher nicht beantwortet

v1 (neueste)

Which of the two programs contains the typical problem of a bottleneck and why? Which rank represents the “bottleneck” here?
Please attach screenshots if you can identify the problem using a visualization in EduMPI.

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

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Frage **13**

Bisher nicht beantwortet

v1 (neueste)

Which of the two broadcast approaches offers better load balancing in an MPI program and why? How do you determine this with EduMPI? Please add screenshots.

Bearbeiten Ansicht Einfügen Format

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Frage **14**

Bisher nicht beantwortet

v1 (neueste)

If you have to distribute a message to all ranks for an MPI program in the future (broadcast), which of the two approaches (collective, p2p) would you choose and why?

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

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◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Frage **15**

Bisher nicht beantwortet

v1 (neueste)

Were you able to answer the questions about the efficiency and runtimes of the broadcast approaches independently or did you need help?

- [ ] I was able to solve the tasks independently.
- [ ] I asked the person sitting next to me or another student for help.
- [ ] I have asked the teacher for help.
- [ ] I looked for help on the Internet or asked ChatGPT.
- [ ] I couldn't do the tasks at all.

Frage **16**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I recognized immediately and without much effort which of the two programs was more efficient.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **17**

Bisher nicht beantwortet

v1 (neueste)

Were you able to solve the tasks for identifying a bottleneck problem independently or did you need help?

- [ ] I was able to solve the tasks independently.
- [ ] I asked the person sitting next to me or another student for help.
- [ ] I have asked the teacher for help.
- [ ] I looked for help on the Internet or asked ChatGPT.
- [ ] I couldn't do the tasks at all.

Frage **18**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

The visualizations in EduMPI have significantly helped to identify a bottleneck problem.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **19**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

EduMPI helped me to better understand the problem of a bottleneck. I now know why I should always avoid bottlenecks in my MPI programs.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **20**

Bisher nicht beantwortet

v1 (neueste)

Did you need help assessing the load balancing of the MPI programs?

- [ ] I was able to solve the tasks independently.
- [ ] I asked the person sitting next to me or another student for help.
- [ ] I have asked the teacher for help.
- [ ] I looked for help on the Internet or asked ChatGPT.
- [ ] I couldn't do the tasks at all.

Frage **21**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

The visualizations in EduMPI did not help to assess the load balancing of the MPI programs

- ( ) Fully applies
- ( ) Applies
- ( ) Partly applies
- ( ) does not apply
- ( ) does not apply at all

◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Information

v1 (neueste)

Now run tests with EduMPI on the cluster using the broadcast_coll.c program.
Use 128 processes for each run.
Edit the size of the array (via the define macro) as follows:

a) ARR_SIZE 100000
b) ARR_SIZE 1000
c) ARR_SIZE 5000

Frage **22**

Bisher nicht beantwortet

v2 (neueste)

Take a close look at the respective program behavior in EduMPI. Describe what you see.
Please use all available views, especially the 3D view with collective communication lines enabled.
Please do not make any interpretations at this point.

Bearbeiten Ansicht Einfügen Format

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Frage **23**

Bisher nicht beantwortet

v2 (neueste)

How is it possible that EduMPI shows different visualizations even though the same program and the same broadcast function are used?
Now try to interpret the analyzed results.

Bearbeiten Ansicht Einfügen Format

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◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Frage **24**

Bisher nicht beantwortet

v1 (neueste)

EduMPI offers several views for viewing (2D, 3D, communication matrix). Which of these views helped best to analyze the programs, understand their behaviour and identify performance issues?

Explain your answer.

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

p 0 Wörter Build with tinyMCE

Frage **25**

Bisher nicht beantwortet

v1 (neueste)

EduMPI offers several options (runtime options) for displaying various communication information in color (send/recv ratio, total send ratio, wait for late sender, etc.). Which option helped you the most to solve the tasks and why?

Bearbeiten Ansicht Einfügen Format

Werkzeuge Tabelle Hilfe

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◀ Usertest 2 - MPI Programs

Direkt zu:

Sie können diesen Test in der Vorschau ansehen. Wäre dies ein realer Versuch, würde dies abgeblockt, weil:

Dieser Test ist momentan nicht verfügbar.

Frage **26**

Bisher nicht beantwortet

v1 (neueste)

Were you able to run the program "broadcast_coll.c" with different sized arrays independently with EduMPI and interpret the results or did you need help?

- [ ] I was able to solve the tasks independently.
- [ ] I asked the person sitting next to me or another student for help.
- [ ] I have asked the teacher for help.
- [ ] I looked for help on the Internet or asked ChatGPT.
- [ ] I couldn't do the tasks at all.

Frage **27**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I was totally confused as to why different performance displays were shown in EduMPI even though I was running the same program.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **28**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

Overall, the user test and EduMPI helped me to better understand the differences between p2p and collective communication. It made me realize why I need collective communication.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **29**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I found the tasks in this user test very difficult and didn't really understand them.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **30**

Bisher nicht beantwortet

v1 (neueste)

To what extent does the following statement apply to you?

I enjoy working with EduMPI and running programs on the cluster.

- Fully applies
- Applies
- Partly applies
- does not apply
- does not apply at all

Frage **31**

Bisher nicht beantwortet

v1 (neueste)

Compared to the last user test, have you discovered new things in EduMPI that you particularly like or dislike? Have you found a bug or want to give new feedback?

Bearbeiten Ansicht Einfügen Format

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◀ Usertest 2 - MPI Programs

Direkt zu: